A POLYTECHNIC INSTITUTION

School of Computing & Academic Studies
Program: Forensic Investigation
Option: Computer Crime Studies

FSCT 8560
Network Security 2

Lab 1: Basic iptables

Part 1: Basic iptables usage

You'll complete this lab on your Ubuntu virtual machine. Follow these steps to get started:

1. Shut down your Ubuntu virtual machine and create a snapshot. After you boot it back up, look at your iptables rules, or lack thereof, by using the following command:

```
sudo iptables -L
```

2. Create the rules that you need for a basic firewall, allowing for Secure Shell access, DNS queries and zone transfers, and the proper types of ICMP. Deny everything else:

```
sudo iptables -A INPUT -m conntrack  --ctstate ESTABLISHED,RELATED  -j ACCEPT

sudo iptables -A INPUT -p tcp --dport ssh -j ACCEPT
sudo iptables -A INPUT -p tcp --dport 53 -j ACCEPT
sudo iptables -A INPUT -p udp --dport 53 -j ACCEPT
sudo iptables -A INPUT -m conntrack -p icmp --icmp-type 3 --ctstate
NEW,ESTABLISHED,RELATED -j ACCEPT

sudo iptables -A INPUT -m conntrack -p icmp --icmp-type 11 --ctstate
NEW,ESTABLISHED,RELATED -j ACCEPT

sudo iptables -A INPUT -m conntrack -p icmp --icmp-type 12 --ctstate
NEW,ESTABLISHED,RELATED -j ACCEPT

sudo iptables -A INPUT -j DROP
```

3. View the results by using the following command:

```
sudo iptables -L
```

4. Add a rule for it at the top of the list for the loopback interface:

```
sudo iptables -I INPUT 1 -i lo -j ACCEPT
```

5. View the results by using the following two commands. Note the difference between the output of each:

```
sudo iptables -L
sudo iptables -L -v
```

6. Install the `iptables-persistent` package and choose to save the IPv4 and IPv6 rules when prompted:

```
sudo apt install iptables-persistent
```

7. Reboot the virtual machine and verify that your rules are still active.

Part 2: Blocking invalid IPv4 packets

For this lab, you'll use the same virtual machine that you used for the previous lab. You won't replace any of the rules that you already have. Rather, you'll just add a couple. Let's get started:

8. Look at the rules for the filter and the mangle tables. (Note that the `-v` option shows you statistics about packets that were blocked by DROP and REJECT rules.) Then, zero out the blocked packets counter:

```
sudo iptables -L -v
sudo iptables -t mangle -L -v
sudo iptables -Z
sudo iptables -t mangle -Z
```

9. From either your host machine or another virtual machine, perform the NULL and Windows Nmap scans against the virtual machine:

```
sudo nmap -sN ip_address_of_your_VM
sudo nmap -sW ip_address_of_your_VM
```

10. Repeat *Step 1*. You should see a large jump in the number of packets that were blocked by the final `DROP` rule in the `INPUT` chain of the filter table:

```
sudo iptables -L -v
sudo iptables -t mangle -L -v
```

11. Make the firewall work more efficiently by using the `PREROUTING` chain of the mangle table to drop invalid packets, such as those that are produced by the two Nmap scans that we just performed. Add the two required rules with the following two commands:

```
sudo iptables -t mangle -A PREROUTING -m conntrack --ctstate INVALID -j DROP

sudo iptables -t mangle -A PREROUTING -p tcp ! --syn -m conntrack --ctstate
NEW -j DROP
```

12. Save the new configuration to your own home directory. Then, copy the file to its proper location and zero out the blocked packet counters:

```
sudo iptables-save > rules.v4
sudo cp rules.v4 /etc/iptables
sudo iptables -Z
sudo iptables -t mangle -Z
```

13. Perform only the NULL scan against the virtual machine:

```
sudo nmap -sN ip_address_of_your_VM
```

14. Look at the iptables ruleset and observe which rule was triggered by the Nmap scan:

```
sudo iptables -L -v
sudo iptables -t mangle -L -v
```

15. This time, perform just the Windows scan against the virtual machine:

```
sudo nmap -sW ip_address_of_your_VM
```

16. Observe which rule was triggered by this scan:

```
sudo iptables -L -v
sudo iptables -t mangle -L -v
```